

Normal measures hidden in critical-sequence quotients

A Prikry factorization of ultraexacting witnesses
and two exact consistency calibrations

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Abstract. The supplied synthesis obtains regularity in parameterized HOD models from rigid classes of cofinal ω -sets modulo finite difference. We develop a stronger conclusion for a critical-sequence class. At an ultraexacting cardinal λ , one can choose a bijection $b : \lambda \rightarrow V_\lambda$ and $q = [a]$ so that the canonical eventual-containment trace

$$U_{b,q} = \{A \in \mathcal{P}(\lambda) \cap \text{HOD}_{\{b,q\}} : a \subseteq^* A\}$$

is a normal measure *inside* $N = \text{HOD}_{\{b,q\}}$, concentrating on the cardinals below λ that are measurable in N . Every representative of q is Prikry-generic over N for this measure. Thus the ambient singularity of λ admits a canonical Prikry subextension with ground model agreeing with the ambient universe through V_λ .

The isolated existence of such a normal trace is equiconsistent with a measurable cardinal. Requiring the trace to concentrate on measurables is equiconsistent with a cardinal of Mitchell order at least two. The upper bounds use a quotient-relative cone-homogeneity argument for ordinary Prikry forcing; the lower bounds use the explicitly defined inner model. These calibrations concern the trace principles, not ultraexactingness itself. Detailed conventional proofs are given, including the required Prikry genericity and homogeneity arguments.

The principal advance

For a suitable quotient-parameter HOD model, the change is

$$\text{regularity} \longrightarrow \text{normal measure} \longrightarrow \text{Prikry genericity}.$$

A further, independent conclusion identifies the precise consistency strength of the two isolated trace principles. No inconsistency of ZFC + I0 or of ultraexacting cardinals is asserted.

Status. Research manuscript with full English proofs; not refereed or formalized in Lean. The new assertions are presented as continuations of the supplied synthesis, not as a claim of established literature priority. Published inputs and the limits of the conclusions are separated explicitly from the arguments developed here.

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1 Statements and their relation to the supplied research

We work in ZFC unless an inner model is explicitly named. A *normal measure on κ* means a nonprincipal, κ -complete ultrafilter on $\mathcal{P}(\kappa)$ with the regressive-function normality property. The words “on $\mathcal{P}(\kappa)$ ” and the completeness quantifier are always evaluated in the model under discussion.

For an uncountable cardinal λ with $\text{cf}^V(\lambda) = \omega$, set

$$D_\lambda = \{a \subseteq \lambda : \text{ot}(a) = \omega \text{ and } \sup a = \lambda\}, \quad Q_\lambda = D_\lambda / \equiv^*,$$

where $a \equiv^* d$ means that $a \Delta d$ is finite. Write $[a]$ for the corresponding set of representatives. In particular, a parameter $q \in Q_\lambda$ is one set parameter; it does not give its individual members as freely available parameters.

For a bijection $b : \lambda \rightarrow V_\lambda$ and $q \in Q_\lambda$, define

$$N_{b,q} = \text{HOD}_{\{b,q\}}^V, \quad (1.1)$$

$$U_{b,q} = \{A \in \mathcal{P}(\lambda) \cap N_{b,q} : (\exists a \in q) a \subseteq^* A\}. \quad (1.2)$$

Here $a \subseteq^* A$ means $a \setminus A$ is finite. The existential quantifier in (1.2) can equally well be universal over q . The fixed-parameter convention and the fact that $N_{b,q}$ satisfies Choice are proved in Section 2.

Definition 1.1 (The two trace principles). The assertion $\text{NT}(\lambda, b, q)$ says that λ is an uncountable cardinal of ambient cofinality ω , that $b : \lambda \rightarrow V_\lambda$ is a bijection, that $q \in Q_\lambda$, and that

$$N_{b,q} \models “U_{b,q} \text{ is a normal measure on } \lambda.”$$

The assertion $\text{NT}^+(\lambda, b, q)$ adds

$$\text{Meas}_\lambda^{N_{b,q}} = \{\alpha < \lambda : N_{b,q} \models “\alpha \text{ is measurable}”\} \in U_{b,q}.$$

Let T_{NT} and T_{NT^+} denote ZFC together with the existence of such a triple for the respective principle.

These are first-order assertions after expanding the definable HOD predicate and relativizing the finitely many formulas about measures. The definition does *not* assert the existence of an ambient normal measure on the singular cardinal λ .

Theorem 1.2 (Ultraexacting normal trace). If λ is ultraexacting, there are b and q such that $\text{NT}^+(\lambda, b, q)$ holds. The class q can be the finite-difference class of the critical sequence of an ultraexacting witness at a sufficiently correct height. For $N = N_{b,q}$ and $U = U_{b,q}$:

1. $V_\lambda^N = V_\lambda$ and N sees λ as measurable, with a normal measure concentrating on measurables.
2. Every $a \in q$, in its increasing enumeration, is $\mathbb{P}_U^N$ -generic over N .
3. $N[a]$ is an inner model of V satisfying

$$V_\lambda^{N[a]} = V_\lambda, \quad \text{cf}^{N[a]}(\lambda) = \omega.$$

Moreover $N[a] = N[d]$ for every $a, d \in q$.

The measure construction is proved in Section 4; the Prikry conclusions are proved in Section 5. No assumption about iterability of an infinite iteration of the witness is used.

Theorem 1.3 (Exact consistency calibrations). The following equivalences of consistency statements hold:

$$\text{Con}(T_{\text{NT}}) \longleftrightarrow \text{Con}(\text{ZFC} + \text{there is a measurable cardinal}), \quad (1.3)$$

$$\text{Con}(T_{\text{NT}^+}) \longleftrightarrow \text{Con}(\text{ZFC} + \exists \kappa \, o(\kappa) \geq 2). \quad (1.4)$$

In the second line, $o(\kappa) \geq 2$ may equivalently be replaced by: there is a normal measure on κ concentrating on measurable cardinals below κ .

The upper bounds are actual forcing constructions, not appeals to a large-cardinal implication diagram. The proofs are in Sections 6–7.

1.1 What has, and has not, been extended

The attached *Synthesis* proves that certain q make λ regular in $\text{HOD}_{V_\lambda \cup \{q\}}$ and develops sharp-width results for definable sections of Q_λ [1]. The present construction uses a special critical-sequence class and a fixed bijection b . Its model $\text{HOD}_{\{b,q\}}$ contains the former parameter-HOD model: each individual parameter in V_λ can be recovered from b and an ordinal. This is a stronger conclusion for a specified model, but is *not* a claim that every rigid class carries the measure in (1.2).

The three assertions in (1.3), (1.4), and “there is an ultraextending cardinal” must be kept distinct. The published equiconsistency of ultraextendingness with I0 remains unchanged [3, Theorem 4.5]. We isolate consequences of that much stronger hypothesis and determine their own exact strengths.

The fixed-witness and canonical-choice methods have precedents in the supplied reports and in the work on exacting cardinals. The continuation here is the conversion of their eventual behavior into normality, the Prikry reconstruction, and the converse quotient-homogeneity construction. Classical Prikry facts are identified and re-proved rather than counted as new results.

2 Fixed-parameter HOD and the canonical trace

2.1 A necessary parameter convention

For a fixed finite tuple p of sets, $\text{OD}_{\{p\}}$ denotes sets definable from p and finitely many ordinal parameters. The tuple may be encoded as one set. Define

$$\text{HOD}_{\{p\}} = \{x : \text{tc}(\{x\}) \subseteq \text{OD}_{\{p\}}\}.$$

This differs from allowing *arbitrary members* of a possibly unwellorderable set as parameters. All HOD models with braces in this paper use only the displayed finite tuple.

Lemma 2.1 (Fixed-parameter inner models). For every fixed finite tuple p , $\text{HOD}_{\{p\}}$ is a definable transitive inner model of ZFC. The class $\text{OD}_{\{p\}}$ has a uniformly definable, set-like canonical well-order. A set of ordinals is in $\text{HOD}_{\{p\}}$ if and only if it is in $\text{OD}_{\{p\}}$.

Proof. A definition can be coded by a rank height, a formula number, and a finite tuple of ordinal parameters. One uses truth in a set structure V_θ containing p , rather than a truth predicate for V . Reflection identifies the resulting definability notion with ordinal definability from p . Order the codes first by a bound on their height and ordinal entries, and then by a fixed well-order within that bound.

Assign each definable object its first code. This produces a uniform set-like well-order of $\text{OD}_{\{p\}}$; in particular every nonempty set of $\text{OD}_{\{p\}}$ objects has a canonical first member.

The hereditary definition gives transitivity, and all ordinals belong to the class. The usual HOD closure argument applies with p held fixed. For example, the internal power set of $x \in \text{HOD}_{\{p\}}$ is $\mathcal{P}(x) \cap \text{HOD}_{\{p\}}$. This is definable from p and a definition of x ; its members and all their descendants are in $\text{OD}_{\{p\}}$. It therefore belongs to $\text{HOD}_{\{p\}}$. Separation and Replacement follow by relativizing a formula to the definable class $\text{HOD}_{\{p\}}$ and substituting definitions for its finitely many parameters. Pairing, Union, Infinity, Foundation, and Extensionality are handled in the same way or by transitivity.

Restrict the canonical order to any set in $\text{HOD}_{\{p\}}$. The resulting well-order is itself hereditarily definable from p , since its ordered pairs are made from hereditarily definable elements. Thus it belongs to the model and proves Choice there. Finally, the descendants of a set of ordinals, other than the set itself, are ordinals. They are already ordinal definable, proving the last assertion. $\square$

The lemma does not say $p \in \text{HOD}_{\{p\}}$. For example, q will be definable from itself but will have no representative in the inner model. Its failure of hereditary definability is essential.

Lemma 2.2 (A rank-bijection supplies the lower universe). Suppose $b : \lambda \rightarrow V_\lambda$ is a bijection and q is any set. Then $N = \text{HOD}_{\{b,q\}}$ contains every member of V_λ , $V_\lambda^N = V_\lambda$, and $b \in N$.

Proof. For $x \in V_\lambda$, choose the ordinal $\xi < \lambda$ with $b(\xi) = x$. Then x is definable from b and ξ . Every member of its transitive closure also lies in V_λ and has the same kind of definition. Consequently $x \in N$. Ranks of sets in transitive inner models are absolute, so this proves the equality of rank segments. The set b itself is definable from the displayed parameters. Its graph elements and all their descendants are in V_λ when λ is a limit cardinal, as in all applications here. Hence $b \in N$. More generally, the same conclusion follows directly by hereditary definability of ordered pairs of ordinals and values of b . $\square$

Lemma 2.3 (The trace is an internal set). For a triple as in Definition 1.1, without assuming normality, $U_{b,q} \in N_{b,q}$. It is a proper filter on $\mathcal{P}(\lambda)^{N_{b,q}}$ containing every final segment of λ .

Proof. Finite modification does not change eventual containment, so the condition in (1.2) is independent of the representative. The set $U_{b,q}$ is definable from b, q and λ , because $N_{b,q}$ is a definable class. Every member A is in $N_{b,q}$ and hence is hereditarily definable from b, q . This shows that the trace itself is hereditarily definable and belongs to $N_{b,q}$.

The empty set does not eventually contain a cofinal ω -set. The whole cardinal does. Eventual containment is upward closed and is preserved by intersections of two sets, giving the filter properties. An increasing ω -sequence cofinal in λ has only finitely many terms below any fixed $\beta < \lambda$: otherwise all its terms would occur below β , contradicting cofinality. Thus every final segment belongs to the trace. $\square$

Normality is a substantial additional assertion. An arbitrary cofinal sequence can alternate between a definable set and its complement, so Lemma 2.3 alone does not even give an ultrafilter.

3 Correct heights, critical sequences, and fixed parameters

3.1 The exact amount of reflection needed

The notions originate in [4]. We use the following form of the definition from [3, Section 2]. At an ultraexacting λ , for arbitrarily high prescribed sufficiently correct cardinal heights ζ , there are

$$X < V_\zeta, \quad V_\lambda \cup \{\lambda\} \subseteq X, \quad j : X \longrightarrow V_\zeta \quad (3.1)$$

with $j(\lambda) = \lambda$, $\text{crit}(j) < \lambda$, and $e = j \upharpoonright V_\lambda \in X$. Here $e : V_\lambda \rightarrow V_\lambda$ is elementary. We choose the height *before* choosing this witness.

Lemma 3.1 (Uniform canonical-counterexample principle). Fix finitely many first-order counterexample predicates, with a finite parameter tuple p and a cardinal parameter λ . Assume their candidate objects have rank below $\lambda + \omega$. There is a finite reflection requirement on V_ζ , uniform in p , with the following consequence. If (3.1) holds, $p \in X$, and $j(p) = p$, then any nonempty class of $\text{OD}_{\{p\}}$ candidates for one of these predicates has a candidate $x \in X$ with $j(x) = x$.

Proof. For each predicate, write a single formula saying that x is its canonical $\text{OD}_{\{p\}}$ -least counterexample. Definability and the canonical order in Lemma 2.1 are first-order definable uniformly in p . The finitely many resulting formulas, their subformulas needed for reflection, and existence and uniqueness of the selected object form a finite list. Choose a cardinal $\zeta > \lambda + \omega$ at which they are correct for all parameters in V_ζ . Equivalently, membership in $C^{(m)}$ for one sufficiently large fixed finite m suffices. Such heights are supplied by Lévy reflection.

If counterexamples exist in V , their canonical first member x has the stipulated bounded rank, so $x \in V_\zeta$. Correctness says that V_ζ selects exactly this same x . Because $p, \lambda \in X$ and $X < V_\zeta$, the unique selected object belongs to X . Elementarity of j , together with $j(p) = p$ and $j(\lambda) = \lambda$, says that $j(x)$ is the same unique object of V_ζ . Thus $j(x) = x$. $\square$

There is no assumption that the ordinal parameters in an arbitrary $\text{OD}_{\{p\}}$ definition are fixed by j . Such an assumption would be false. The canonical least object is fixed because it is uniquely definable from the *fixed tuple*, not because its original defining code is pointwise fixed. Uniform reflection also avoids choosing a height only after the witness has supplied b and q .

3.2 The critical sequence and its lower-rank measures

Lemma 3.2 (Critical sequence facts). For a witness (3.1) let

$$\kappa = \text{crit}(j) = \text{crit}(e), \quad \kappa_n = e^n(\kappa), \quad a = \{\kappa_n : n < \omega\}.$$

Then every κ_n is measurable in V , $\sup_{n < \omega} \kappa_n = \lambda$, and λ is a strong limit cardinal of cofinality ω . The enumeration of a and the set a are in X , and

$$j(a) = a \setminus \{\kappa\}.$$

Consequently $q = [a] \in X$ and $j(q) = q$.

Proof. First recall the derived-measure argument. Set

$$U_0 = \{A \subseteq \kappa : \kappa \in e(A)\}.$$

All relevant subsets, functions, and sequences below κ have rank below λ , so e acts on them. It follows immediately from complements and finite intersections that U_0 is an ultrafilter. Every singleton is null, since ordinals below κ are fixed. If $\mu < \kappa$ and $\langle A_i : i < \mu \rangle$ consists of members of U_0 , then e fixes the index set and sends its intersection to the intersection of the images. Thus κ belongs to that image and the intersection lies in U_0 .

For a regressive $f : S \rightarrow \kappa$ with $S \in U_0$, we have $e(f)(\kappa) = \beta < \kappa$. The ordinal β is fixed, so $\kappa \in e(\{\alpha \in S : f(\alpha) = \beta\})$. This proves normality. The critical point is uncountable, since finite ordinals and ω are fixed. Hence U_0 witnesses measurability. The statement that it is a normal measure is absolute between V_λ and V , all its quantifiers ranging over sets of bounded rank here. Applying e^n to U_0 therefore gives a normal measure on κ_n in V .

We use the classical rank form of Kunen's inconsistency: there is no nontrivial elementary self-embedding of $V_{\delta+2}$ in ZFC [5]. If $\delta = \sup_n \kappa_n < \lambda$, the critical sequence has bounded rank in V_λ , and $e(\delta) = \delta$. Restricting e to $V_{\delta+2}$ would give just such a forbidden embedding. Thus $\delta = \lambda$. Measurable cardinals are strongly inaccessible, so λ is a countable increasing limit of strong limits and is itself strong limit. For completeness, regularity of a measurable follows by ruling out a partition into fewer than κ bounded null sets; strong limit follows by taking a purported injection $\kappa \rightarrow \mathcal{P}(\theta)$ for $\theta < \kappa$ and intersecting the measure-one decisions of all θ membership coordinates. The resulting measure-one fiber would have at most one element.

The recursion defining $\langle \kappa_n : n < \omega \rangle$ is a set recursion in V_ζ from e . Since $e \in X$, elementarity places its unique value in X . A function with domain ω is mapped pointwise by j , and $j(\kappa_n) = e(\kappa_n) = \kappa_{n+1}$. This proves the stated formula for $j(a)$. Forming the finite-difference class is definable from a, λ , and is correctly computed at this height. Deleting one element does not change the class, so $q \in X$ is fixed. $\square$

Only the explicitly stated classical Kunen theorem is imported in this argument. No assertion about an iterable direct limit of the iterates of e is required.

Lemma 3.3 (A fixed bijection). The witness in Lemma 3.2 supplies a bijection $b : \lambda \rightarrow V_\lambda$ with $b \in X$ and $j(b) = b$.

Proof. As κ is inaccessible, choose a bijection $b_0 : \kappa \rightarrow V_\kappa$. Its rank is below λ , so $b_0 \in V_\lambda \subseteq X$. Define

$$b_n = e^n(b_0) : \kappa_n \longrightarrow V_{\kappa_n}.$$

Every ordinal below κ and every member of V_κ is fixed by e . Therefore $b_0 \subseteq e(b_0)$. Applying e repeatedly gives $b_n \subseteq b_{n+1}$. Consequently

$$b = \bigcup_{n < \omega} b_n$$

is a bijection from λ to V_λ .

The sequence of the b_n is uniquely obtained by recursion from e and b_0 , both in X , so the sequence and its union belong to X . Each $b_n \in V_\lambda$, hence $j(b_n) = e(b_n) = b_{n+1}$. Mapping the countable sequence pointwise and taking unions gives

$$j(b) = \bigcup_{n < \omega} b_{n+1} = b.$$

This is the bijection version of the fixed-well-order device also used in [3, Proposition 3.9] and in the supplied synthesis. $\square$

4 From eventual decisions to a normal measure

Fix the height in Lemma 3.1 for the two predicates used below, then choose the ultraexacting witness and its a, q, b . Write $p = (b, q)$, $N = \text{HOD}_{\{p\}}$, and $U = U_{b,q}$. Every parameter in p is fixed by j .

Lemma 4.1 (Eventual decisions). For every $A \subseteq \lambda$ in $\text{OD}_{\{p\}}$, either $a \subseteq^* A$ or $a \subseteq^* (\lambda \setminus A)$.

Proof. Suppose otherwise. A counterexample is an $\text{OD}_{\{p\}}$ subset of λ that splits a representative of q into two infinite pieces. This property is independent of the representative, is first-order in p, λ , and has candidates of bounded rank. By Lemma 3.1 there is such a counterexample $A_0 \in X$ with $j(A_0) = A_0$.

For every $n < \omega$, elementarity gives

$$\kappa_n \in A_0 \iff j(\kappa_n) = \kappa_{n+1} \in j(A_0) = A_0.$$

Membership in A_0 is therefore constant along the entire critical sequence. Thus either $a \subseteq A_0$ or $a \cap A_0 = \emptyset$, contradicting the choice of A_0 . $\square$

Lemma 4.2 (Eventual pressing down). Let $f \in \text{OD}_{\{p\}}$ have domain $S \subseteq \lambda \setminus \{0\}$ and satisfy $f(\alpha) < \alpha$ for $\alpha \in S$. If $a \subseteq^* S$, then f is eventually constant along a .

Proof. Failure is again a first-order counterexample predicate in p, λ : the graph is a set of ordinal pairs of rank below $\lambda + \omega$; its domain eventually contains a representative of q ; and there is no ordinal β with that representative eventually contained in the fiber $f^{-1}\{\beta\}$. Select a canonical counterexample $f_0 \in X$ with $j(f_0) = f_0$ by Lemma 3.1.

The set $S_0 = \text{dom}(f_0)$ is fixed. Membership of κ_n in S_0 is constant in n by the same calculation as in Lemma 4.1. Since membership holds eventually, it holds for every n , including $n = 0$. Put $\beta = f_0(\kappa)$. Regressiveness gives $\beta < \kappa$, so $j(\beta) = \beta$. Elementarity of function evaluation now yields, successively,

$$f_0(\kappa_{n+1}) = j(f_0(\kappa_n)) = \beta.$$

The function is constant on all of a , a contradiction. $\square$

Proposition 4.3 (Internal normality and completeness). The set U belongs to N and N regards it as a normal measure on λ .

Proof. Membership $U \in N$ and the proper filter properties were proved in Lemma 2.3. Every $A \in \mathcal{P}(\lambda)^N$ is in $\text{OD}_{\{p\}}$, so Lemma 4.1 decides it or its complement. These two alternatives cannot both occur, because a is infinite. Thus U is an ultrafilter on the full internal power set. It contains all final segments and no singleton, so it is nonprincipal.

If $S \in U$ and $f \in N$ is regressive on $S \setminus \{0\}$, then $f \in \text{OD}_{\{p\}}$ and $a \subseteq^* S$. Lemma 4.2 gives an ordinal β such that $a \subseteq^* f^{-1}\{\beta\}$. The fiber is in N and in U . This is the regressive-function normality property for all such f in N .

We verify λ -completeness rather than presupposing it. Let $\langle A_i : i < \mu \rangle \in N$, where $\mu < \lambda$ and each $A_i \in U$. Its intersection belongs to N . Suppose the intersection were not in U . The ultrafilter property would put its complement in U . After intersecting with the final segment above μ , obtain $B \in U$ and define in N

$$f(\alpha) = \min\{i < \mu : \alpha \notin A_i\} \quad (\alpha \in B).$$

This is regressive. Normality gives an $i < \mu$ whose fiber is in U . But that fiber is disjoint from $A_i \in U$, contradicting the proper filter property. The case $\mu = 0$ is immediate. This proves completeness for every sequence of length less than λ that belongs to N . $\square$

Proposition 4.4 (Concentration on measurables). For $M_\lambda = \text{Meas}_\lambda^N$, we have $a \subseteq M_\lambda$ and $M_\lambda \in U$. In particular $\text{NT}^+(\lambda, b, q)$ holds.

Proof. The model N contains V_λ by Lemma 2.2. For each n , the normal measure on κ_n obtained in Lemma 3.2 has rank below λ , and hence belongs to N . All subsets of κ_n and all functions needed to test its normality and completeness also have rank below λ . Thus the same measure witnesses measurability in N .

The set M_λ is defined by Separation inside N and consists of ordinals, so it is an eligible argument of the trace. It contains every term of a , proving $M_\lambda \in U$ directly from the definition. Together with Proposition 4.3, this establishes the measure part of Theorem 1.2. $\square$

Where the additional strength enters

Eventual decisions give an ultrafilter. Eventual pressing down gives normality. A regressive “least failed coordinate” then supplies internal λ -completeness. Merely knowing that λ is regular in a parameterized HOD model would not supply any of these three steps.

5 Prikry reconstruction, with the genericity proof

The results in this section about normal measures and Prikry forcing are classical. The genericity criterion is due to Mathias [6]; see also the formulation in [7]. We include proofs because the fact that our inner model need not be countable makes a mere “meet its dense sets one by one” argument inappropriate.

5.1 The forcing and finite-stem diagonal intersections

Temporarily work inside a transitive model $M \models \text{ZFC}$ with a normal measure W on κ . A condition in $\mathbb{P}_W$ is a pair (s, A) , where s is a finite increasing sequence of ordinals below κ , $A \in W$, and every element of A exceeds every entry of s . Empty stems are allowed. The order is

$$(t, B) \leq (s, A) \iff \begin{cases} s \text{ is an initial segment of } t, \\ \text{all new entries of } t \text{ lie in } A, \\ B \subseteq A. \end{cases}$$

A direct extension, written $\leq^*$, has the same stem. For a finite sequence t , write A/t for the elements of A above every entry of t ; thus $A/\emptyset = A$.

Lemma 5.1 (Diagonalization over finite stems). If $\langle A_s : s \in [\kappa]^{<\omega} \rangle \in M$ and each $A_s \in W$, then

$$\Delta_{\text{fin}} A_s = \{ \beta < \kappa : (\forall s \in [\beta]^{<\omega}) \beta \in A_s \} \in W.$$

Finite subsets and their increasing enumerations are identified here.

Proof. All choices and functions in this proof are taken in M . Suppose the complement were in W . Intersect it with $A_\emptyset$ and with $\kappa \setminus \{0\}$. For each remaining β , choose a nonempty finite $s_\beta \subseteq \beta$ with $\beta \notin A_{s_\beta}$. The function $\beta \mapsto \max(s_\beta)$ is regressive. Normality makes it constant, with value γ , on a set in W .

There are fewer than κ finite subsets of $\gamma + 1$. By κ -completeness and the ultrafilter property, on a further set in W the whole finite set s_β has a constant value s . That set is disjoint from $A_s \in W$, a contradiction. The intersection with $A_\emptyset$ also accounts for the empty-stem case. $\square$

Lemma 5.2 (Finite homogeneity). For every $n < \omega$, every $\rho < \kappa$, and every coloring $h : [\kappa]^n \rightarrow \rho$ in M , there is $H \in W$ homogeneous for h .

Proof. For $n = 0$ there is only one input. For $n = 1$, a partition into fewer than κ color classes has exactly one class in W . Suppose the result holds for n , and consider a coloring of $(n + 1)$ -element sets. For each $s \in [\kappa]^n$, there is a color $i_s < \rho$ such that

$$B_s = \{\beta > \max s : h(s \cup \{\beta\}) = i_s\} \in W.$$

For $n = 0$ the maximum restriction is omitted. Existence of i_s again uses κ -completeness. Apply the induction hypothesis to the coloring $s \mapsto i_s$ and obtain a homogeneous $H_0 \in W$. Intersect H_0 with the finite-stem diagonal intersection of the B_s (assign κ to unused stem lengths). On the resulting set H , every increasing $(n + 1)$ -tuple has its last entry in the appropriate B_s and its first n entries in H_0 . Its color is therefore the single color fixed by the induction hypothesis. $\square$

Lemma 5.3 (Strong Prikry lemma). If $D \in M$ is dense open in $\mathbb{P}_W$ and (s, A_0) is a condition, there are $A \subseteq A_0$ in W and $n < \omega$ such that

$$(s \smallfrown t, A/t) \in D \quad \text{for every } t \in [A]^n.$$

Proof. For every finite increasing t from A_0 , color t good if there exists $B \in W$ such that $(s \smallfrown t, B) \in D$, and bad otherwise. For each length separately, Lemma 5.2 makes this coloring constant on a set in W . Intersect these countably many sets with A_0 . Since κ is uncountable and W is κ -complete, the result H still belongs to W .

By density, some $(s \smallfrown t, B) \leq (s, H)$ belongs to D . Thus the homogeneous color at length $n = |t|$ is good. For each $u \in [H]^n$ choose a witness $B_u \in W$ with $(s \smallfrown u, B_u) \in D$. Apply Lemma 5.1 to these reservoirs, assigning κ to all unused stems, and let A be the intersection of that diagonal set with H . If $u \in [A]^n$ and $\beta \in A/u$, then $u \subseteq \beta$, so $\beta \in B_u$. Hence $A/u \subseteq B_u$. The condition $(s \smallfrown u, A/u)$ extends the chosen condition in D , and belongs to D because D is open. This includes $n = 0$, using the empty-stem reservoir in the diagonal intersection. $\square$

5.2 Mathias's criterion without a countability hypothesis

Let $c = \langle c_n : n < \omega \rangle$ be strictly increasing and cofinal in κ in an outer universe containing M . Every finite initial segment of c belongs to M , since it is a finite sequence of ordinals. Put

$$G_c = \{(s, A) \in \mathbb{P}_W^M : s \text{ is an initial segment of } c \text{ and all remaining entries of } c \text{ lie in } A\}. \quad (5.1)$$

Theorem 5.4 (Mathias genericity criterion). The filter G_c is M -generic if and only if

$$(\forall A \in W) (\exists k < \omega) (\forall n \geq k) c_n \in A. \quad (5.2)$$

Proof. First, G_c is a filter whenever (5.2) holds. It is upward closed in the forcing order. For two conditions in it, their stems are comparable initial segments of c . Take the longer stem and intersect the two reservoirs above that stem. The intersection belongs to W , contains the remaining tail of c , and

gives a common extension in G_c . Nonemptiness is immediate, for example from the empty stem with reservoir κ .

Assume (5.2), and fix a dense open $D \in M$. For every finite stem s , apply Lemma 5.3 below the condition with stem s and full reservoir above it. By Choice in M , choose simultaneously reservoirs $A_s \in W$ and integers n_s such that all n_s -point stem extensions from A_s , with their residual reservoir, belong to D . Form $A = \Delta_{\text{fin}} A_s \in W$ in M .

Choose k so that $c_n \in A$ for all $n \geq k$, and set $s = c \restriction k$. For $n \geq k$, every entry of s is less than c_n ; therefore $c_n \in A_s$ by the definition of the diagonal set. Let t be the next n_s entries of c . The strong Prikry conclusion gives

$$(s \frown t, A_s/t) \in D.$$

All the still later entries of c lie in this reservoir, so the condition belongs to G_c . This proves $G_c \cap D \neq \emptyset$. No enumeration of the dense subsets of the forcing was used.

Conversely, for any $A \in W$, conditions whose reservoirs are subsets of A form a dense set: intersect the current reservoir with A . A generic filter meets that set. Beyond the corresponding stem, every entry of its Prikry sequence is in A . This proves (5.2). $\square$

Corollary 5.5 (The Prikry subextension of a trace). Suppose $\text{NT}(\lambda, b, q)$ holds, and write $N = N_{b,q}$, $U = U_{b,q}$. Then every representative $a \in q$ is Prikry-generic over N . The model $N[a]$ satisfies ZFC, lies inside V , has $V_\lambda^{N[a]} = V_\lambda$, and sees $\text{cf}(\lambda) = \omega$. For $a, d \in q$, $N[a] = N[d]$.

Proof. By definition of the trace, the increasing enumeration of a is eventually in every member of U . Apply Theorem 5.4 inside N , with this external sequence. Evaluation of N -names by the filter (5.1) takes place in V , giving a forcing extension $N[a] \subseteq V$ satisfying ZFC. It contains a cofinal increasing ω -sequence in λ .

The lower rank equality follows especially directly here: $V_\lambda \subseteq N \subseteq N[a] \subseteq V$. Transitivity and rank absoluteness give equality at height λ .

If $a \neq^* d$, their finite symmetric difference is a finite set of ordinals, and so belongs to N . The operation of taking symmetric difference shows $d \in N[a]$ and $a \in N[d]$. These mutual containments imply equality of the two forcing extensions. $\square$

This completes Theorem 1.2. It is useful to write $W_{b,q}$ for the common model $N_{b,q}[a]$, $a \in q$. This notation names a Prikry subextension, not a claim that the full ambient universe is $W_{b,q}$.

5.3 Preservation facts needed for the converse construction

We record the remaining classical facts with their proofs so that the upper consistency bounds do not hide a preservation assumption.

Proposition 5.6 (Prikry property and no bounded subsets). Every sentence in the forcing language is decided by a direct extension of any given condition. Consequently $\mathbb{P}_W$ adds no new subsets of any ordinal below κ .

Proof. Apply the strong Prikry lemma to the dense open set of conditions deciding a fixed sentence φ . It gives (s, A) and a length n such that every canonical n -point stem extension decides φ . Color these n -element stems by their two possible decisions. Lemma 5.2 gives $B \subseteq A$ in W on which the decision is constant. Any condition below (s, B) can be extended to have at least n new stem entries. It then

extends one of the canonical deciding conditions, and so makes the constant decision. The conditions making that decision are dense below (s, B) ; hence (s, B) itself decides φ . Its stem is still s .

Now let $\dot{x}$ be a name forced to be a subset of $\alpha < \kappa$. Starting from any condition, use direct extensions to decide $\xi \in \dot{x}$ successively for all $\xi < \alpha$. At limit stages, intersect the previous reservoirs. There are fewer than κ stages, so κ -completeness gives a reservoir in W , while the stem never changes. The final direct extension decides all of $\dot{x}$ as one ground-model subset of α . Such conditions are dense, so every generic extension has no new bounded subsets of κ . $\square$

Proposition 5.7 (Cardinals and the lower rank segment). If G is $\mathbb{P}_W$ -generic over M , then in $M[G]$ all cardinals of M remain cardinals,

$$\text{cf}(\kappa) = \omega, \quad V_\kappa^{M[G]} = V_\kappa^M.$$

Proof. The union of generic stems is increasing. Dense sets requiring a longer stem make its order type ω , and dense sets requiring an entry above any specified $\beta < \kappa$ make it cofinal.

The no-bounded-subsets result preserves cardinals below κ : a new collapse between two smaller ordinals could be coded by a new subset of an ordinal below κ , using a ground-model pairing bijection. The cardinal κ itself remains a cardinal because it was a limit of smaller cardinals, all still cardinals. A collapse of κ to an ordinal of size below it would also collapse one of those intervening cardinals.

There are κ possible finite stems, and two conditions with the same stem are compatible by intersecting their reservoirs. Thus the forcing is κ^+ -cc. The usual antichain argument preserves all cardinals at least κ^+ : for a name for a function with a given domain, each argument has at most κ possible ordinal values. A domain of ground-model cardinality ν can therefore cover at most $\nu \cdot \kappa$ possible values, which cannot be a larger ground cardinal.

Finally, κ is inaccessible in M . Induct on $\alpha < \kappa$. Assume the two models have the same V_α . A ground-model bijection of this set with an ordinal below κ codes any of its subsets by a bounded subset of κ . There are no new such codes, so the models have the same $V_{\alpha+1}$. The limit step is union. This proves equality at κ as well. $\square$

6 Quotient-relative homogeneity and realization of traces

We now reverse the construction. Start in a model $M \models \text{ZFC}$ with a normal measure W on a measurable κ . Let G be generic for $\mathbb{P}_W^M$, let c be its increasing Prikry sequence, and work in

$$V^* = M[G], \quad q = [\text{ran}(c)].$$

Choose in M a bijection $b : \kappa \rightarrow V_\kappa^M$. This remains a bijection onto $V_\kappa^{V^*}$ by Proposition 5.7. The quotient q in this section is computed in the extension V^* .

6.1 Stem replacement fixes the quotient, not the sequence

Lemma 6.1 (Isomorphic cones with the same quotient). For any two conditions (s, A) and (t, B) there are direct extensions (s, C) and (t, C) whose cones are isomorphic by

$$(s \smallfrown u, D) \mapsto (t \smallfrown u, D).$$

This isomorphism fixes ground-model parameters and preserves the canonical name for the finite-difference class of the Prikry sequence.

Proof. Let $C = A \cap B$ restricted above every entry of s and t . It belongs to W . An extension below (s, C) consists exactly of a finite new stem u from C and a reservoir $D \subseteq C$ above the whole new stem; replacing the old prefix s by t gives precisely an extension below (t, C) . The inverse replaces t by s , and the order is preserved.

Paired generics for these cones give sequences $s \frown d$ and $t \frown d$ with the same infinite tail d . Their ranges differ by a finite set. They also generate the same extension of M : either sequence is recovered from the other by replacing a ground-model finite prefix. In this common extension the two finite-difference classes are literally the same set. Thus, under the induced translation of names, the names for q are forced equal. Check-names for ground-model sets, including b and all ordinal parameters, are fixed in the usual sense of cone isomorphisms.

This argument is a forcing-name argument and does not require actual generics over the entire ambient universe. One can equivalently carry it out in the Boolean completions of the two cones. $\square$

Lemma 6.2 (HOD containment despite the generic parameter). In V^* ,

$$K = \text{HOD}_{\{b,q\}}^{V^*} \subseteq M.$$

Proof. First consider any sentence with parameters $\check{b}$, the canonical name $\check{q}$, and check-names for ordinal parameters. It cannot be forced true by one condition and false by another: shrink to the isomorphic cones from Lemma 6.1; their isomorphism preserves the sentence and would carry a forcing of truth to a forcing of truth below a condition forcing falsity. Since deciding conditions are dense, the top condition decides every such sentence.

Suppose $x \subseteq \theta$ is ordinal definable from b, q in V^* . Fix a defining formula and its finite ordinal parameter tuple. For each $\alpha < \theta$, the assertion that α belongs to the uniquely defined set is a sentence of the preceding form. In M , use the forcing relation to collect exactly the α for which the top condition forces this assertion. Separation in M produces a set $x_0 \subseteq \theta$, and the truth lemma shows $x = x_0$. Thus every $\text{OD}_{\{b,q\}}^{V^*}$ set of ordinals belongs to M .

To pass from sets of ordinals to hereditary definability, let $x \in K$. In V^* , put the canonical $\text{OD}_{\{b,q\}}$ order on $\text{tc}(\{x\})$ and enumerate it in that order by an ordinal η . Code the membership relation of this enumeration as a relation E on η , and code E as a set of ordinals using a fixed ordinal-pair coding. The code is ordinal definable from b, q , so by the preceding paragraph $E \in M$. It is well-founded and extensional. The well-foundedness witnessed in V^* implies the well-foundedness that M needs to perform its Mostowski collapse. The collapse is absolute and unique, and its value at the ordinal corresponding to x is x . Hence $x \in M$. $\square$

The hereditary qualification is indispensable. The generic quotient q is itself definable from q , but it does not follow that $q \in M$. The argument establishes containment for HOD, not for the entire class of ordinal-definable objects from the quotient parameter.

6.2 The old measure is recovered on the new inner power set

Theorem 6.3 (Prikry realization of a normal trace). In the above extension V^* , the triple (κ, b, q) satisfies NT. For $K = \text{HOD}_{\{b,q\}}^{V^*}$ its canonical trace is

$$U_{b,q} = W \cap \mathcal{P}(\kappa)^K. \quad (6.1)$$

If W concentrates in M on measurable cardinals below κ , then (κ, b, q) satisfies NT^+ in V^* .

Proof. The ambient cardinal and cofinality conditions follow from Proposition 5.7. Lemma 2.2 gives $V_\kappa^{V^*} \subseteq K$, whereas Lemma 6.2 gives $K \subseteq M$. In particular these models agree on V_κ .

For any $A \in \mathcal{P}(\kappa)^K$, we have $A \in M$. If $A \in W$, the Prikry sequence is eventually in A , by genericity and the reservoir density argument. If $A \notin W$, its complement belongs to W , so the sequence is eventually outside A . Therefore

$$\text{ran}(c) \subseteq^* A \iff A \in W.$$

This proves (6.1). Independently of that equality, Lemma 2.3 puts the left side, as a set, inside K . This is important: it is not being assumed that the original measure W itself belongs to K .

The right side of (6.1) is a normal measure in K . Indeed, a sequence in K of length $\mu < \kappa$ of its members is a sequence in M of members of W . Its intersection belongs to W by completeness in M , and belongs to K by the axioms of K , so it belongs to the restricted measure. A regressive function in K is also in M . Normality of W produces a constant fiber in W ; its ordinal value and the fiber belong to K , so normality holds for the restriction. The ultrafilter and nonprincipal properties follow at once from those of W and equality of complements. Hence NT holds.

Finally, measurability of a cardinal $\alpha < \kappa$ is absolute among K, M, V^* , because their V_κ segments agree. Every measure witness on α , every subset of α , and every sequence or regressive function used to test it has rank below κ . Thus

$$\text{Meas}_\kappa^K = \text{Meas}_\kappa^M.$$

If the latter set belongs to W , then it belongs to the trace by (6.1). This proves NT⁺. $\square$

Remark 6.4 (No definability assumption on the ground measure). The construction works with an arbitrary normal measure in M . We did not require $M = \text{HOD}$, W ordinal definable, or a definable well-order of the whole universe. The ground-model bijection b gives all lower-rank objects to K , and the quotient q defines the restriction of W to K .

7 The two equiconsistency theorems

7.1 Concentration and Mitchell height two

For normal measures W_0, W_1 on κ , write

$$W_0 \triangleleft W_1 \iff W_0 \in \text{Ult}(V, W_1),$$

where the ultrapower is identified with its transitive collapse. The two-step form of “ $o(\kappa) \geq 2$ ” is the existence of $W_0 \triangleleft W_1$. This explicit formulation suffices throughout and avoids any need for higher Mitchell-rank calculations.

Lemma 7.1 (The concentration criterion). For a normal measure W on κ in a universe of ZFC, the following are equivalent:

1. $\{\alpha < \kappa : \alpha \text{ is measurable}\} \in W$.
2. There is a normal measure W_0 on κ with $W_0 \in \text{Ult}(V, W)$.

Thus existence of a measure as in (1) is equivalent to existence of a cardinal of Mitchell order at least two.

Proof. Let $i : V \rightarrow M_W$ be the normal ultrapower. Normality gives $[\text{id}]_W = \kappa$: every ordinal predecessor represented by a regressive function is equal modulo W to a constant below κ . By Łoś's theorem,

$$\{\alpha < \kappa : \alpha \text{ is measurable}\} \in W \iff M_W \models \text{"}\kappa \text{ is measurable."} \quad (7.1)$$

We verify that a normal measure witnessing the right side is also a normal measure in the original universe. For every $A \subseteq \kappa$ in V ,

$$A = i(A) \cap \kappa \in M_W.$$

Consequently M_W and V have exactly the same subsets of κ . They also have the same relations on κ and the same sequences of subsets of κ of length at most κ : such a relation or sequence is coded by a subset of $\kappa \times \kappa$, and a pairing bijection in M_W converts that code to a subset of κ .

If M_W has a normal measure W_0 on κ , it therefore measures all the ambient subsets, not a smaller power set. Every ambient regressive function and every sequence of fewer than κ measure-one sets used in testing normality or completeness is already in M_W . The internal verification of these properties is thus an ambient verification. This proves (1) $\Rightarrow$ (2).

Conversely, if an ambient normal measure W_0 belongs to M_W , then it is a normal measure there as well: the power sets agree, and the internal tests are among the ambient tests. Hence the right side of (7.1) holds, proving (2) $\Rightarrow$ (1). $\square$

When this lemma is used in $N_{b,q}$, every occurrence of the universe and ultrapower in it is interpreted in $N_{b,q}$. It does not turn the trace into a measure on the full ambient power set.

7.2 Measurable strength for the basic trace

Proof of (1.3). For the lower bound, a model of T_{NT} has the explicitly definable inner model $N_{b,q}$. By Lemma 2.1 this model satisfies ZFC, and the defining property of NT supplies its measurable cardinal λ . Therefore consistency of T_{NT} implies consistency of ZFC with a measurable cardinal.

For the upper bound, start with ZFC and a measurable κ , choose a normal measure W , and force with $\mathbb{P}_W$. Choose a ground-model bijection $b : \kappa \rightarrow V_\kappa$ and let q be the finite-difference class of the generic sequence. Theorem 6.3 proves that the extension satisfies $\text{NT}(\kappa, b, q)$. Hence consistency of the measurable-cardinal theory implies consistency of T_{NT} . $\square$

Corollary 7.2 (The stronger clause is not automatic for a trace). The construction of Theorem 6.3 can be performed at the first measurable cardinal of its ground model. At that particular κ , $K = \text{HOD}_{\{b,q\}}$ regards κ as its first measurable cardinal. Thus its trace is normal but does not concentrate on measurables below κ .

Proof. The models K and the ground agree below κ . No smaller cardinal has a measure in the ground, and hence none does in K . The canonical trace makes κ measurable in K . The set of smaller measurables is therefore empty and is not in the trace. $\square$

This is a statement about the constructed triple. It is not a claim that the entire resulting universe has no other possible NT^+ triple, which would require an additional global argument.

7.3 Mitchell height two for the strengthened trace

Proof of (1.4). For the lower bound, a model of T_{NT^+} has the inner model $N_{b,q} \models \text{ZFC}$ with a normal measure on λ concentrating on its measurable cardinals. Apply Lemma 7.1 inside that model. It sees $o(\lambda) \geq 2$, establishing the required consistency implication.

For the upper bound, assume there are normal measures $W_0 \triangleleft W_1$ on κ . By Lemma 7.1, W_1 concentrates on measurables below κ . Force with $\mathbb{P}_{W_1}$, choose the bijection b in the ground, and form the generic quotient q in the extension. The last clause of Theorem 6.3 gives $NT^+(\kappa, b, q)$ there. $\square$

7.4 Why these are ordinary consistency statements

The model notation in the forcing proofs is the customary way to present a relative consistency argument; it does not add the hypothesis that a countable transitive model exists. The lower bounds are uniform inner-model interpretations. For each axiom of ZFC, its relativization to the fixed-parameter HOD predicate is provable in ZFC, uniformly in the parameters. The measure assertions likewise relativize by explicit first-order formulas.

For the upper bounds, ZFC with the stated measure hypothesis proves that the appropriate set forcing forces every axiom of ZFC and the respective first-order trace assertion. The forcing theorem, or its Boolean-valued presentation, converts these proofs into the ordinary syntactic relative consistency implication. Consequently (1.3) and (1.4) concern $\text{Con}(T)$, not the stronger assertion $\text{Con}(\text{there is a transitive model of } T)$.

8 The HOD boundary and conditional incompatibilities

8.1 Eliminating the rank-bijection parameter

Suppose the additional boundary condition $V_\lambda \subseteq \text{HOD}^V$ holds. For $q \in Q_\lambda$ put

$$N_q^0 = \text{HOD}_{\{q\}}^V, \quad U_q^0 = \{A \in \mathcal{P}(\lambda) \cap N_q^0 : (\exists a \in q) a \subseteq^* A\}.$$

The superscript 0 distinguishes this model from the synthesis's $\text{HOD}_{V_\lambda \cup \{q\}}$ and from $N_{b,q}$.

Theorem 8.1 (A quotient-only normal measure at the boundary). Assume λ is ultraextending and $V_\lambda \subseteq \text{HOD}^V$. There is a critical-sequence class q such that N_q^0 satisfies ZFC, has $V_\lambda^{N_q^0} = V_\lambda$, and regards U_q^0 as a normal measure on λ concentrating on measurables. Every representative of q is Prikry-generic over N_q^0 for U_q^0 , and all give the same Prikry subextension of V .

Proof. Repeat the height choice and critical-sequence construction, now using only $p = q$ in the two counterexample predicates of Lemmas 4.1 and 4.2. The quotient is still fixed by j . Those two proofs use no property of b beyond its being a fixed parameter, so they give eventual decisions and eventual pressing down for all $\text{OD}_{\{q\}}$ subsets and functions.

Lemma 2.1 gives ZFC for N_q^0 . The boundary condition supplies $V_\lambda \subseteq \text{HOD} \subseteq N_q^0$, replacing the use of Lemma 2.2. The proof of Proposition 4.3 therefore gives the normal measure, and the lower-rank measures on the critical points give concentration as in Proposition 4.4. Apply the Mathias criterion and the finite-modification argument in Corollary 5.5 to obtain all the remaining conclusions. $\square$

Corollary 8.2 (An enhanced I0-equiconsistent boundary package). The following theory is equiconsistent with ZFC + I0: there is an ultraexacting λ with $V_\lambda \subseteq \text{HOD}$, no exacting cardinal below λ , and a quotient-only normal trace and Prikry subextension as in Theorem 8.1.

Proof. The published results [3, Theorem 4.5, Proposition 3.9 and Corollary 3.10] supply, relative to I0, an ultraexacting cardinal in a model with the displayed HOD boundary and no smaller exacting cardinal. Theorem 8.1 supplies the new normal trace and Prikry clauses in that same model. Conversely, the stated theory includes an ultraexacting cardinal, so its consistency implies consistency of I0 by the published equiconsistency theorem. $\square$

Unlike Theorem 1.3, this last equiconsistency *inherits* its two large-cardinal bounds from the literature. Its new content is the stronger structural conclusion in the boundary model, not a new comparison between I0 and ultraexactingness.

8.2 A precise conditional inconsistency

Corollary 8.3 (Low-measure HOD hypotheses are incompatible with ultraexactingness). There is no ultraexacting λ satisfying the following additional hypothesis:

For every bijection $b : \lambda \rightarrow V_\lambda$ and every $q \in Q_\lambda$, $\text{HOD}_{\{b,q\}}$ has no normal measure on λ concentrating on its measurable cardinals below λ .

Equivalently, one may require all these models to have internal $o(\lambda) \leq 1$ (with $o(\lambda) = 0$ when there is no normal measure). At the HOD boundary, it suffices to quantify over $\text{HOD}_{\{q\}}$, without b .

Proof. Theorem 1.2 contradicts the displayed hypothesis at its specific pair (b, q) ; Lemma 7.1, interpreted internally, gives the equivalent Mitchell-order formulation. Under the boundary condition, use Theorem 8.1 instead. $\square$

This is an explicit additional hypothesis about parameterized inner models. It is not a theorem that this hypothesis holds in ZFC, nor an inconsistency of a standard large-cardinal axiom by itself. The converse forcing constructions show why normal traces cannot be rejected simply because their supporting sequences are countable in the ambient universe.

9 Internal completeness, omitted information, and proof audit

9.1 The external countable intersection that does fail

Proposition 9.1 (Exactly where ambient completeness fails). In the conclusion of Theorem 1.2, set

$$B_n = \lambda \setminus (\kappa_n + 1) \quad (n < \omega).$$

Then every $B_n \in U_{b,q}$ but $\bigcap_{n < \omega} B_n = \emptyset$ in V . The sequence $\langle B_n : n < \omega \rangle$ does not belong to $N_{b,q}$. Moreover $q \notin N_{b,q}$ and $q \cap N_{b,q} = \emptyset$.

Proof. Every B_n is a final segment, belongs to the inner model by its ordinal definition, and eventually contains the critical sequence. The intersection is empty because that sequence is cofinal in λ . If the

sequence of these sets were in $N_{b,q}$, internal countable completeness would put the empty intersection in the measure, contradicting propriety. Thus the sequence itself is missing from the inner model.

Inside $N_{b,q}$, the cardinal λ is measurable and therefore regular. A representative $d \in q \cap N_{b,q}$ would have order type ω and supremum λ there as well: its increasing enumeration and order type are absolute between the transitive models. That would contradict regularity. Hence there is no such representative. If q itself belonged to the transitive model, so would all of its members, which is impossible. $\square$

There are two separate restrictions on the trace. Its domain is the *internal* power set $\mathcal{P}(\lambda)^{N_{b,q}}$, and its completeness applies to sequences *in* that model. Neither can be replaced by its ambient version. In particular, the trace cannot be inserted as an ambient complete ultrafilter into the HCD arguments of the supplied completeness-barrier analysis.

The parameter q can define the measure without belonging to the hereditarily definable model. This is not a conflict with the HOD construction: U has only hereditarily definable descendants, whereas q has non-hereditarily-definable representatives among its members.

9.2 A ground-model restriction, not a ground-model identification

Proposition 9.2 (A small chain-condition presentation is impossible). For $N = N_{b,q}$ as in Theorem 1.2, V is not a forcing extension of N by a forcing that is λ -cc in N .

Proof. Suppose such a forcing supplied a countable cofinal sequence in λ . For each coordinate, choose in N a maximal antichain deciding its value. Each antichain has cardinality less than λ in N . Their countable union has size less than λ in N , since N regards λ as regular uncountable. The set of all possible values is therefore bounded below λ in N , and the same ordinal bound holds in V , a contradiction. $\square$

This is the familiar regularity obstruction, now applied to the normal-trace model. Our positive construction only proves $N \subseteq N[a] \subseteq V$. It gives no forcing presentation of V over N or over $N[a]$. Ordinary Prikry forcing is λ^+ -cc, not λ -cc, so the positive and negative statements are compatible.

9.3 Dependency and failure-mode audit

Step	What is used, and what is not assumed
Correct height	A finite list of first-order predicates is reflected uniformly in the later parameters. No full truth predicate for V or post hoc choice of a height for a previously fixed witness is used.
Fixed objects	$e \in X$ puts the critical recursion and the union of iterated bijections in X . Arbitrary elements of X are not assumed fixed.
Canonical choice	A least OD_p counterexample is uniquely definable from the fixed tuple. The ordinal parameters in its original definition need not be fixed.
Normality	The value of a fixed regressive function at the actual critical point is below that point and hence fixed. This argument does not apply to arbitrary orbit roots.

Step	What is used, and what is not assumed
Completeness	The least-failed-coordinate function is formed from a sequence that belongs to the inner model. External sequences are excluded, as Proposition 9.1 demonstrates.
Prikry genericity	The normal-measure diagonalization meets any one inner-model dense set. No countability of the inner model is assumed.
Homogeneity	Stem replacement preserves the entire finite-difference class. It does not preserve a distinguished generic representative.
Restriction of measure	The ground measure need not be in the HOD model. Its restriction is there because it equals the canonically definable trace.
Mitchell order	The two-step calculation is performed in the named universe. In the ultraexacting application it is an internal $N_{b,q}$ calculation.
Consistency	The two trace calibrations use inner-model interpretations and set forcing. They do not use a transitive-model existence hypothesis.
Formal verification	No new Lean proof or Lean compilation is claimed. The supplied Lean files were reference material, not a certificate for the new normality or forcing arguments.

9.4 Questions that genuinely remain after this continuation

The work does not settle whether ordinary exactingness rules out thin definable complete sections, whether more than λ rigid classes must exist, or whether a cover-exacting cardinal can consistently lie above a strongly compact cardinal. In particular, the external failure of completeness prevents using the present measure as a shortcut through the last question.

A focused new question is whether every rigid quotient class admits *some* canonical normal or extender trace, perhaps after a change of the quotient presentation. The proof here does not establish this: eventual pressing down uses the first point of the critical sequence and its position below the critical point's image. Rigidity alone records cardinalities of definable families and does not encode that regressive-function calculation.

Another concrete question is whether the Prikry subextension $W_{b,q}$ can be characterized without the chosen fixed bijection b away from the HOD boundary. The boundary theorem removes b under a specific hypothesis, but no independence-of- b theorem is proved in the general case. These questions retain the structural information exposed by the normal trace rather than trying to infer an ambient measure on a singular cardinal.

10 Concluding perspective

The critical sequence has two simultaneous interpretations. In V it is a countable cofinal sequence through a singular strong limit. In its fixed-parameter HOD model, its finite-difference class determines which internal subsets are eventually accepted, and the embedding forces this acceptance rule to be a normal measure. Mathias's criterion then identifies the same sequence as Prikry-generic over that model.

The converse construction explains the exact strength of the isolated phenomenon. Finite stem changes are invisible to the quotient parameter, so parameter-HOD stays inside a measurable ground model. The generic quotient nevertheless reconstructs the old measure on the inner power set. Adding concentration on measurables raises this isolated consistency strength from a measurable cardinal to Mitchell height two, and no further for the stated principle.

Thus the new conclusion is not an ambient contradiction. It is an explicit normal-measure/Prikry structure beneath the ambient singularity, together with two exact consistency calibrations and a sharper conditional obstruction to uniformly low-measure parameterized HOD models.

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